```
Mozambique, step 2 (2026-10-20)

 chain     seed iterations.saved acceptance.rate final.log.likelihood best.log.likelihood errored.proposals
     1 20263016             5000           0.277             -33892.5            -33878.0                 0
     2 20263017             5000           0.277             -33889.7            -33879.8                 0
     3 20263018             5000           0.276             -33893.3            -33881.3                 0
     4 20263019             5000           0.275             -33895.3            -33879.6                 0
                                          file
 mcmc_mozambique_step2_chain1_2026-10-19.Rdata
 mcmc_mozambique_step2_chain2_2026-10-20.Rdata
 mcmc_mozambique_step2_chain3_2026-10-20.Rdata
 mcmc_mozambique_step2_chain4_2026-10-20.Rdata

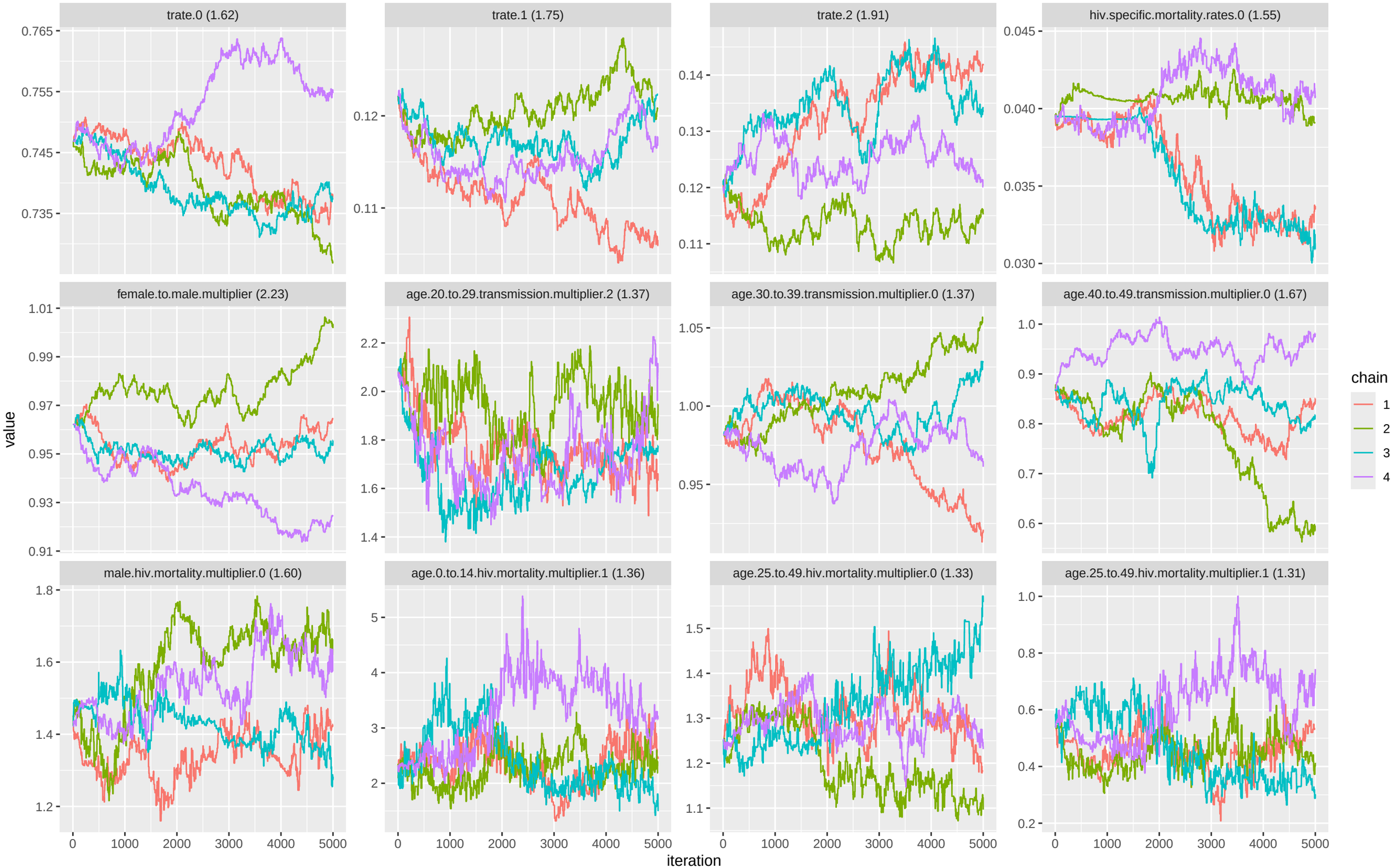
Worst R-hats (> 1.1 suggests the chains haven't converged):
              female.to.male.multiplier                                trate.2                                trate.1
                                   2.231                                  1.912                                  1.747
 age.40.to.49.transmission.multiplier.0                                trate.0        male.hiv.mortality.multiplier.0
                                   1.665                                  1.617                                  1.602
         hiv.specific.mortality.rates.0 age.20.to.29.transmission.multiplier.2 age.30.to.39.transmission.multiplier.0
                                   1.553                                  1.368                                  1.365
 age.0.to.14.hiv.mortality.multiplier.1 age.25.to.49.hiv.mortality.multiplier.0 age.25.to.49.hiv.mortality.multiplier.1
                                   1.360                                  1.332                                  1.312
 age.15.to.19.transmission.multiplier.0 age.30.to.39.transmission.multiplier.2 age.40.to.49.transmission.multiplier.1
                                   1.292                                  1.281                                  1.273
 age.20.to.29.transmission.multiplier.0 age.20.to.29.transmission.multiplier.1         hiv.specific.mortality.rates.1
                                   1.271                                  1.247                                  1.247
     over.50.hiv.mortality.multiplier.1                                trate.4                                trate.3
                                   1.203                                  1.178                                  1.163
 age.15.to.19.transmission.multiplier.2                    age.assortativity         hiv.specific.mortality.rates.2
                                   1.163                                  1.149                                  1.149
 age.30.to.39.transmission.multiplier.1
                                   1.141

Merged simset:
   location simulations total.weight                                                  file
 Mozambique        1000         1000 simset_mozambique_step2_chains1234_2026-10-20.Rdata
```

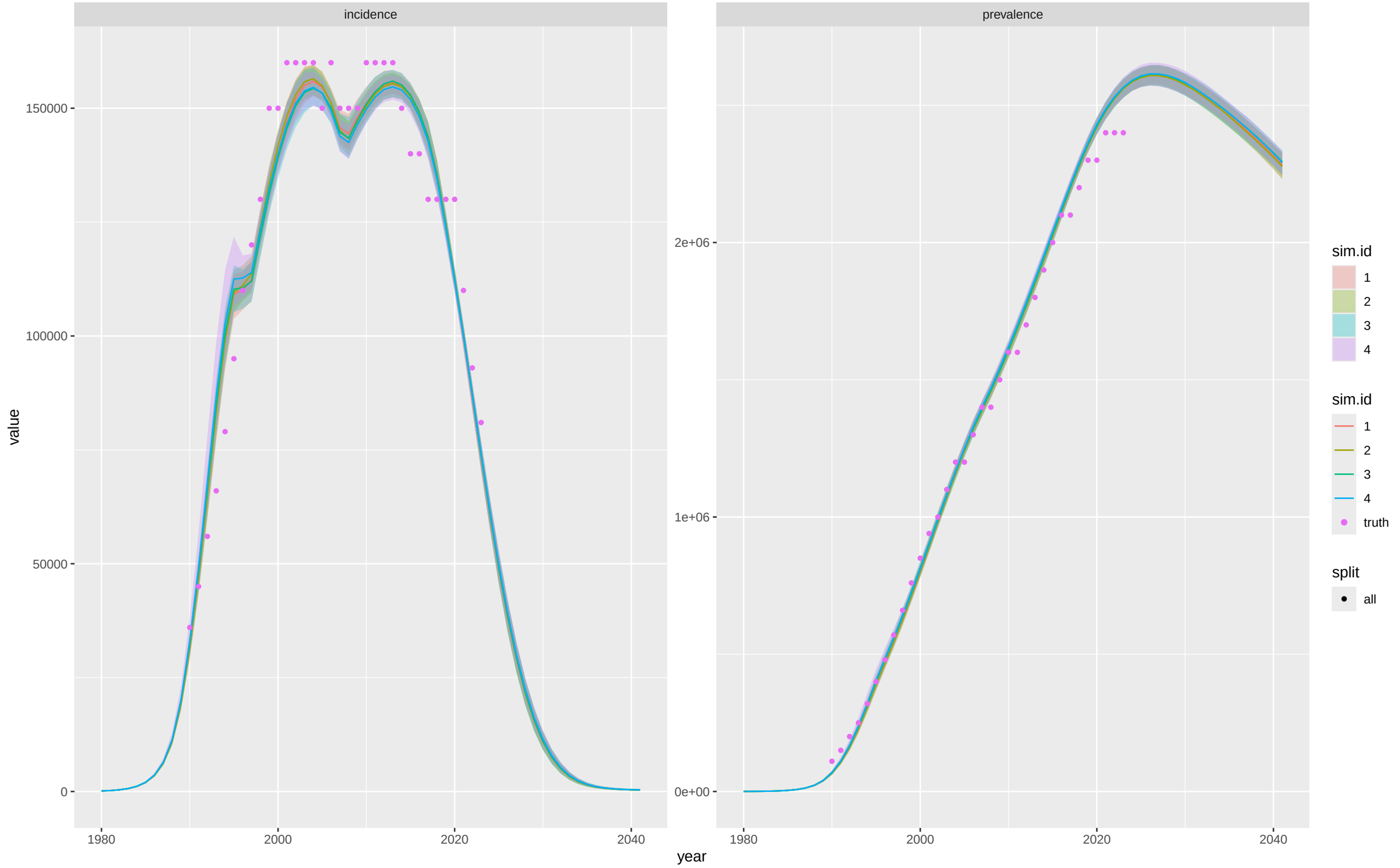
Mozambique step 2: likelihood

0
−10000
−20000
−30000
value
0
25000
50000
75000
100000
iteration
chain
1
2
3
4
type
Log Likelihood
Log Prior
Log Prior x Likelihood

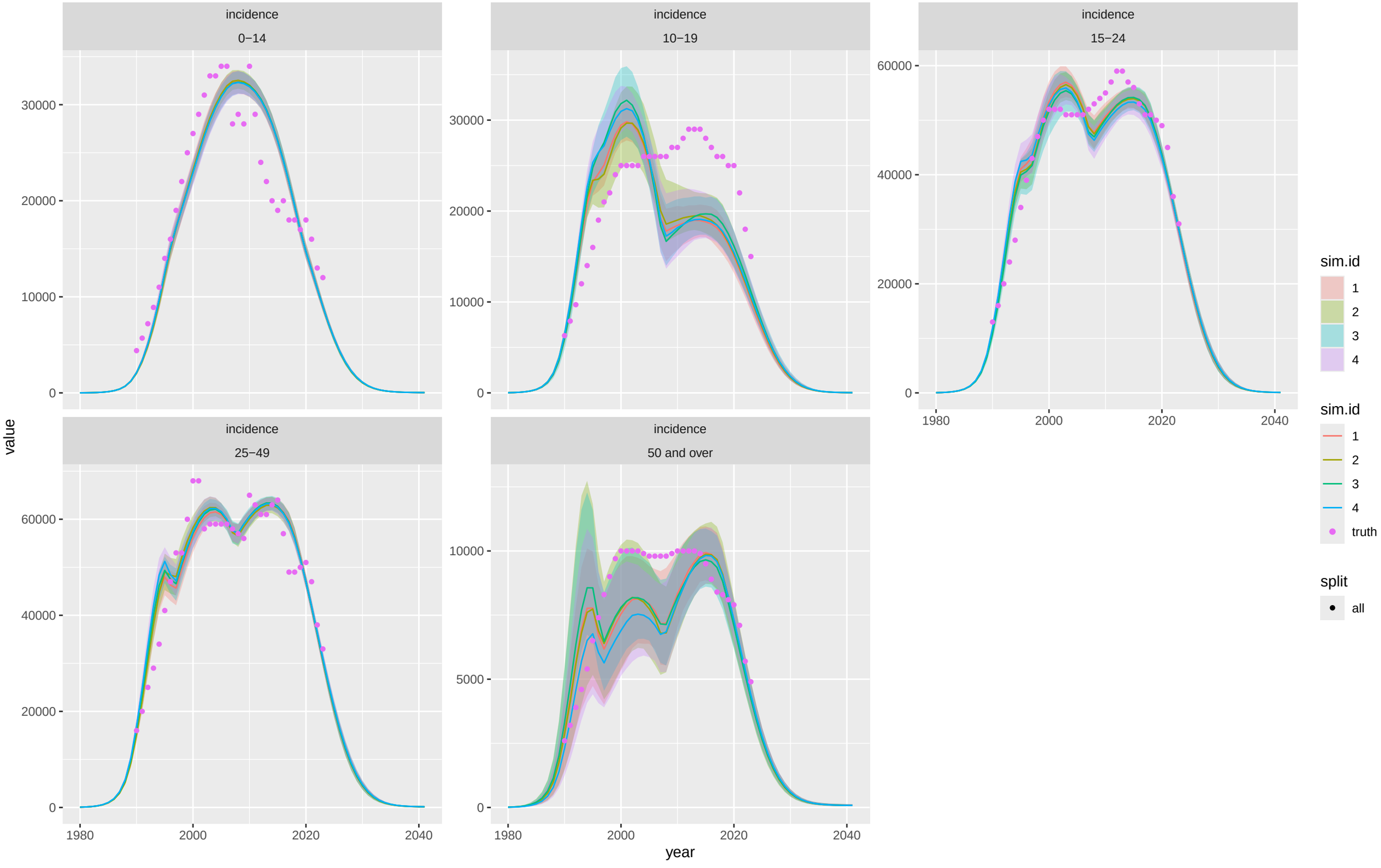
Mozambique step 2: trace



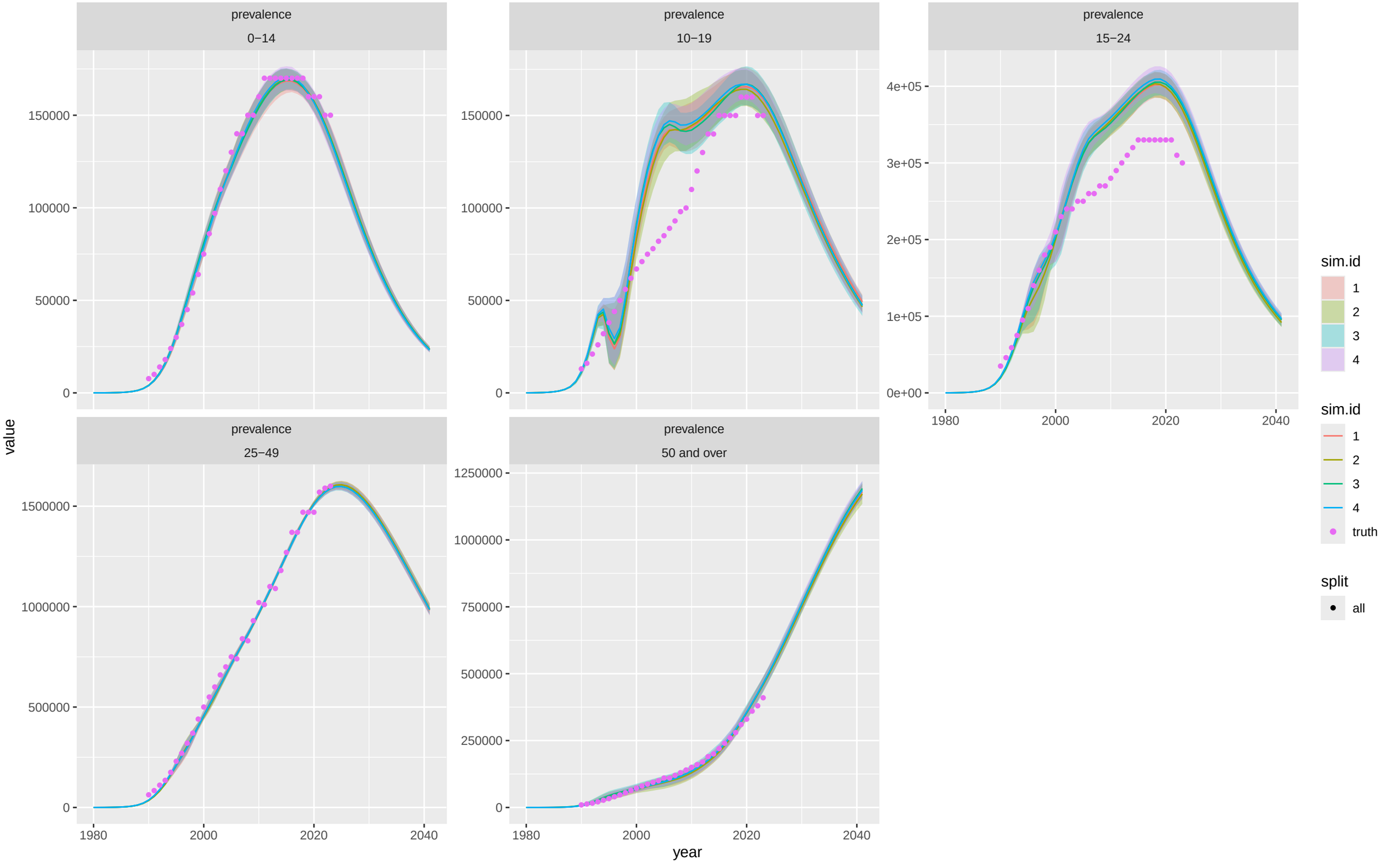
Mozambique step 2: fit (one color per chain)



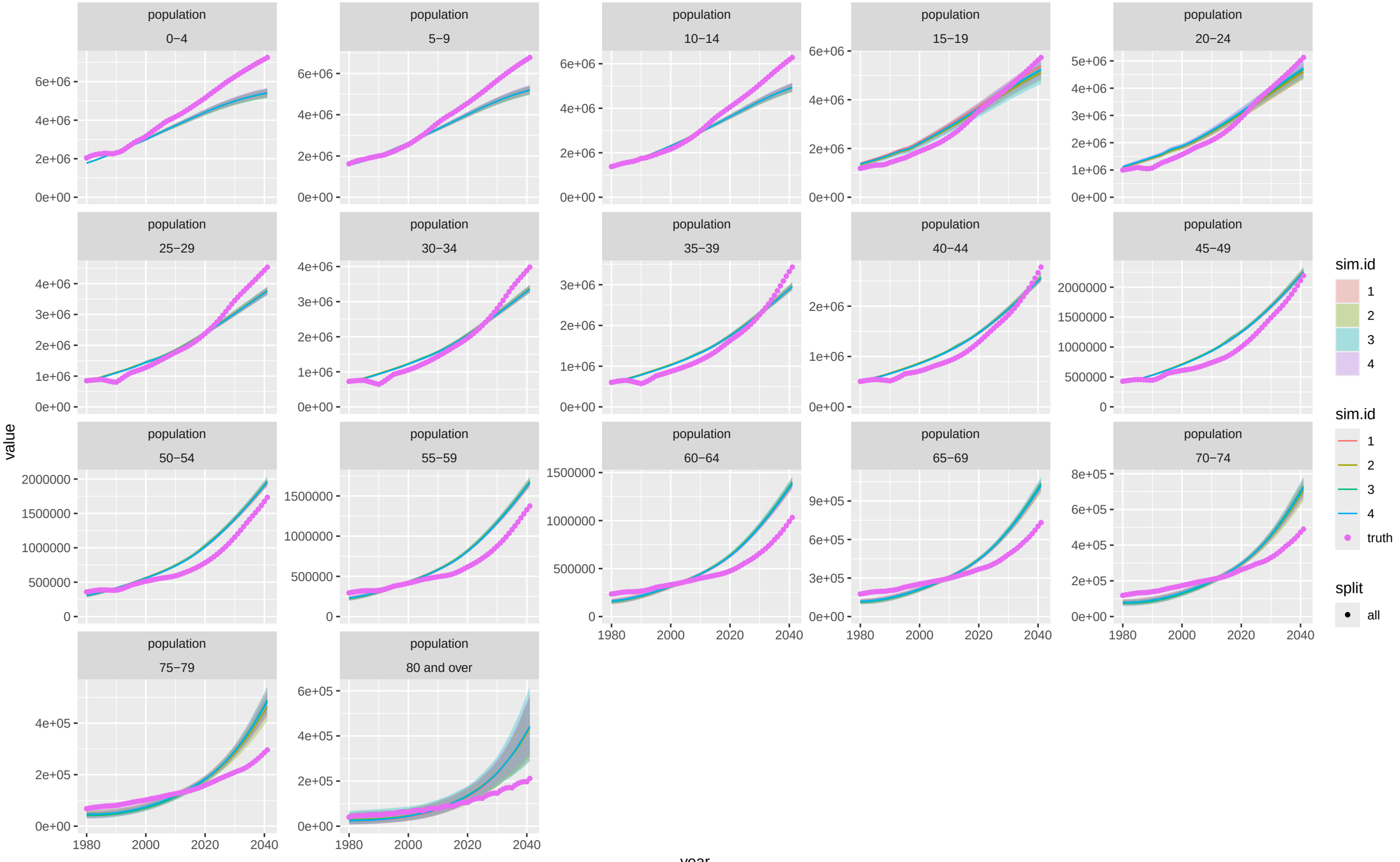
Mozambique step 2: incidence.by.age (one color per chain)



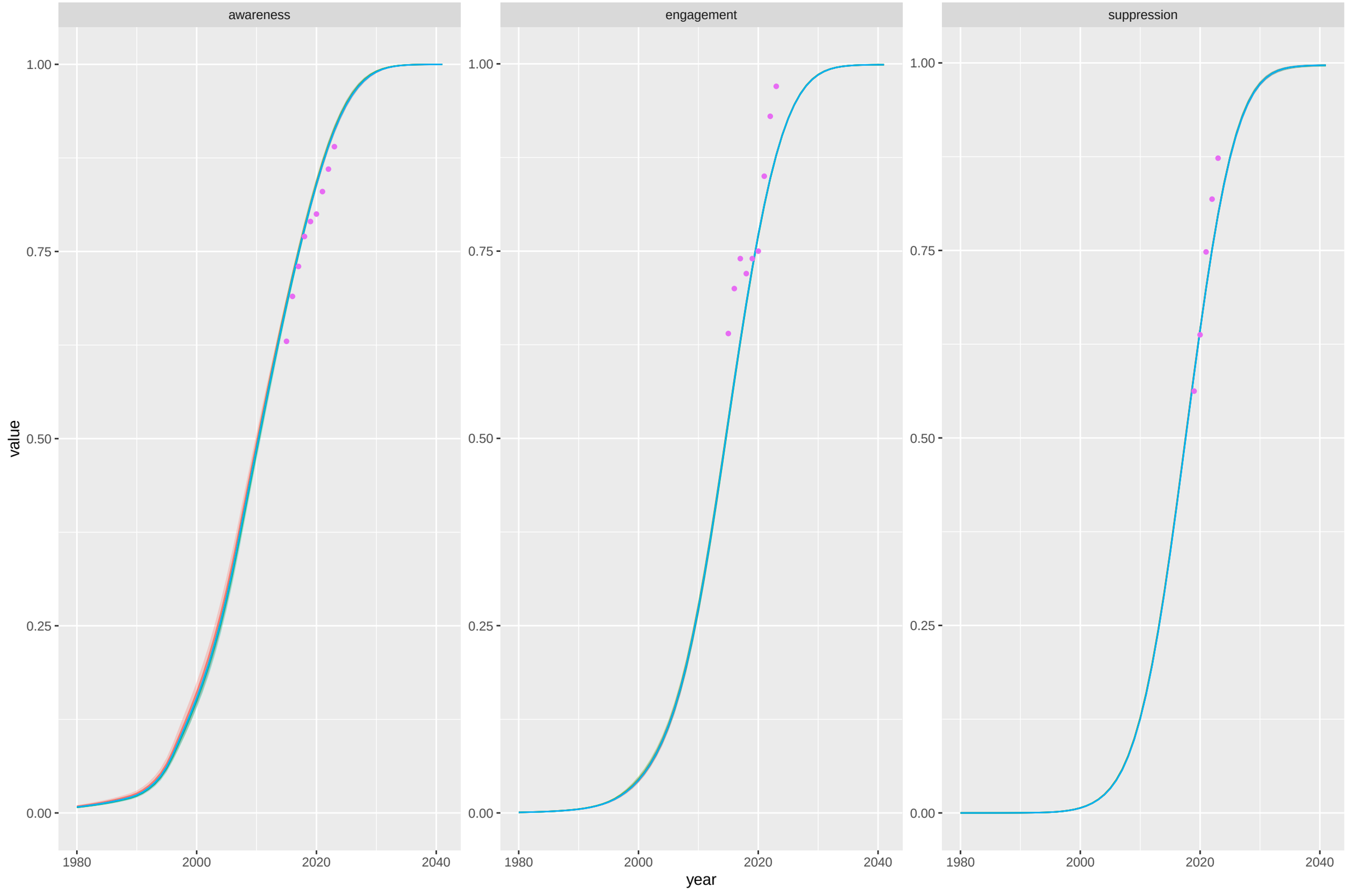
Mozambique step 2: prevalence.by.age (one color per chain)



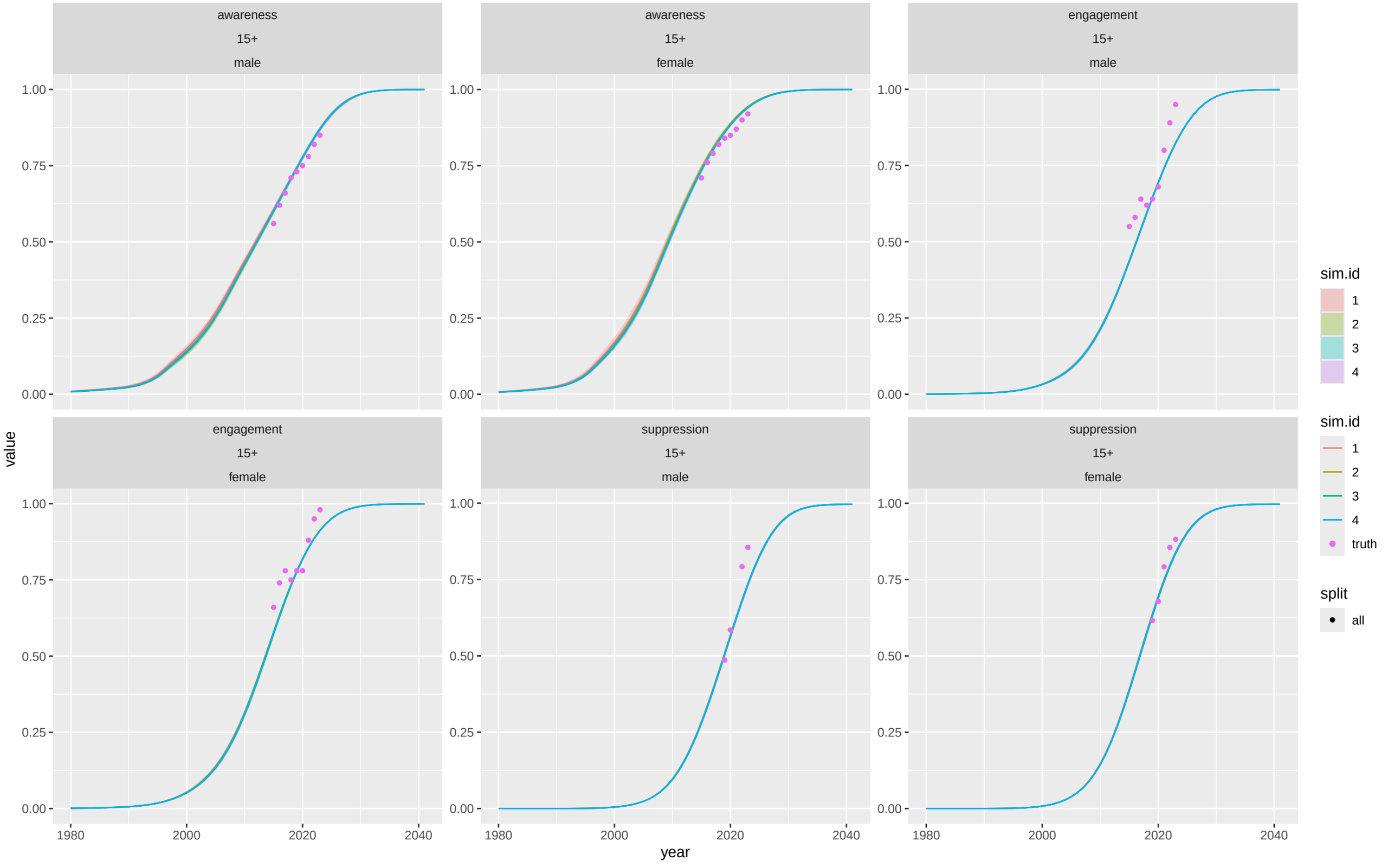
Mozambique step 2: population.by.age (one color per chain)



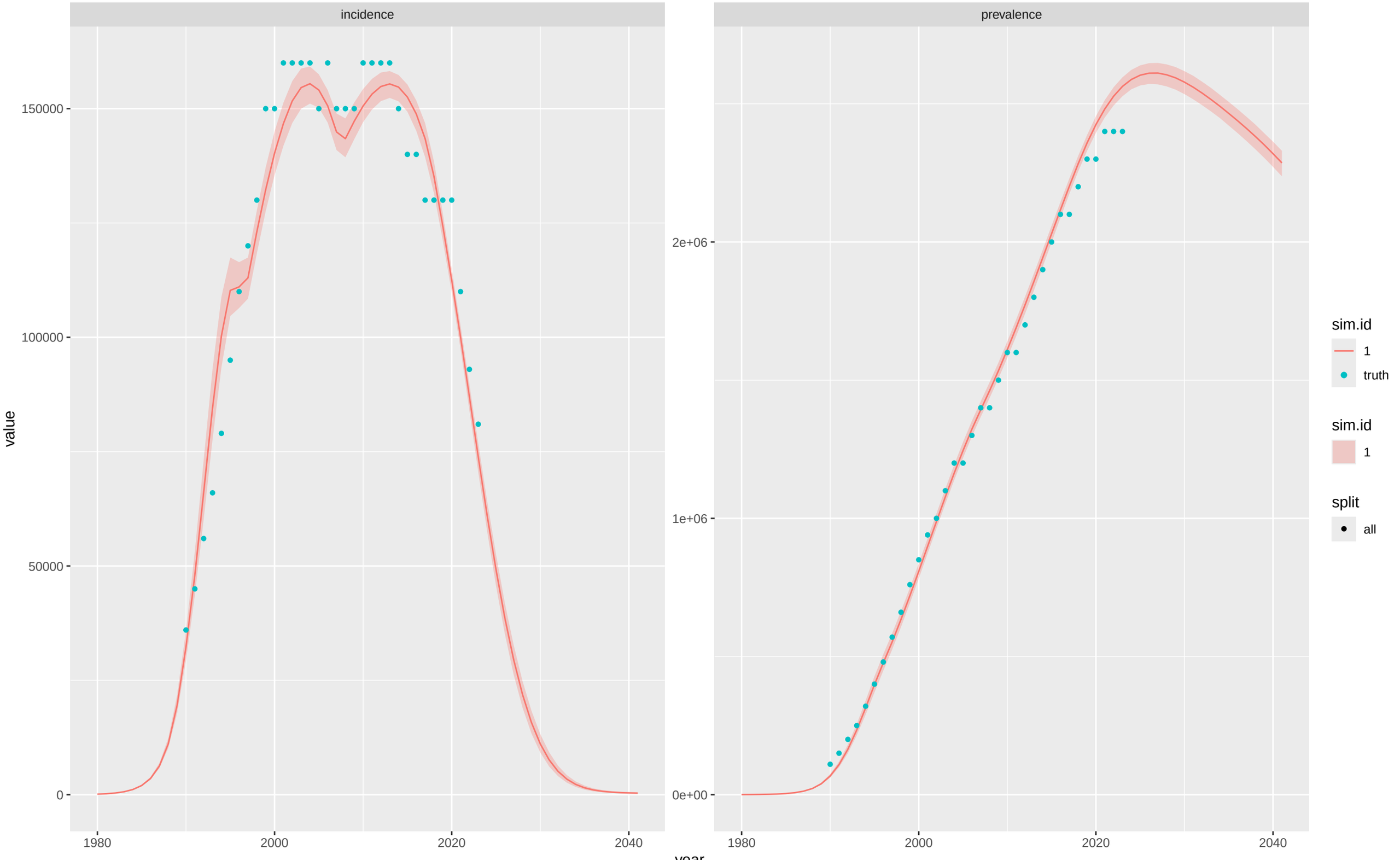
Mozambique step 2: cascade (one color per chain)



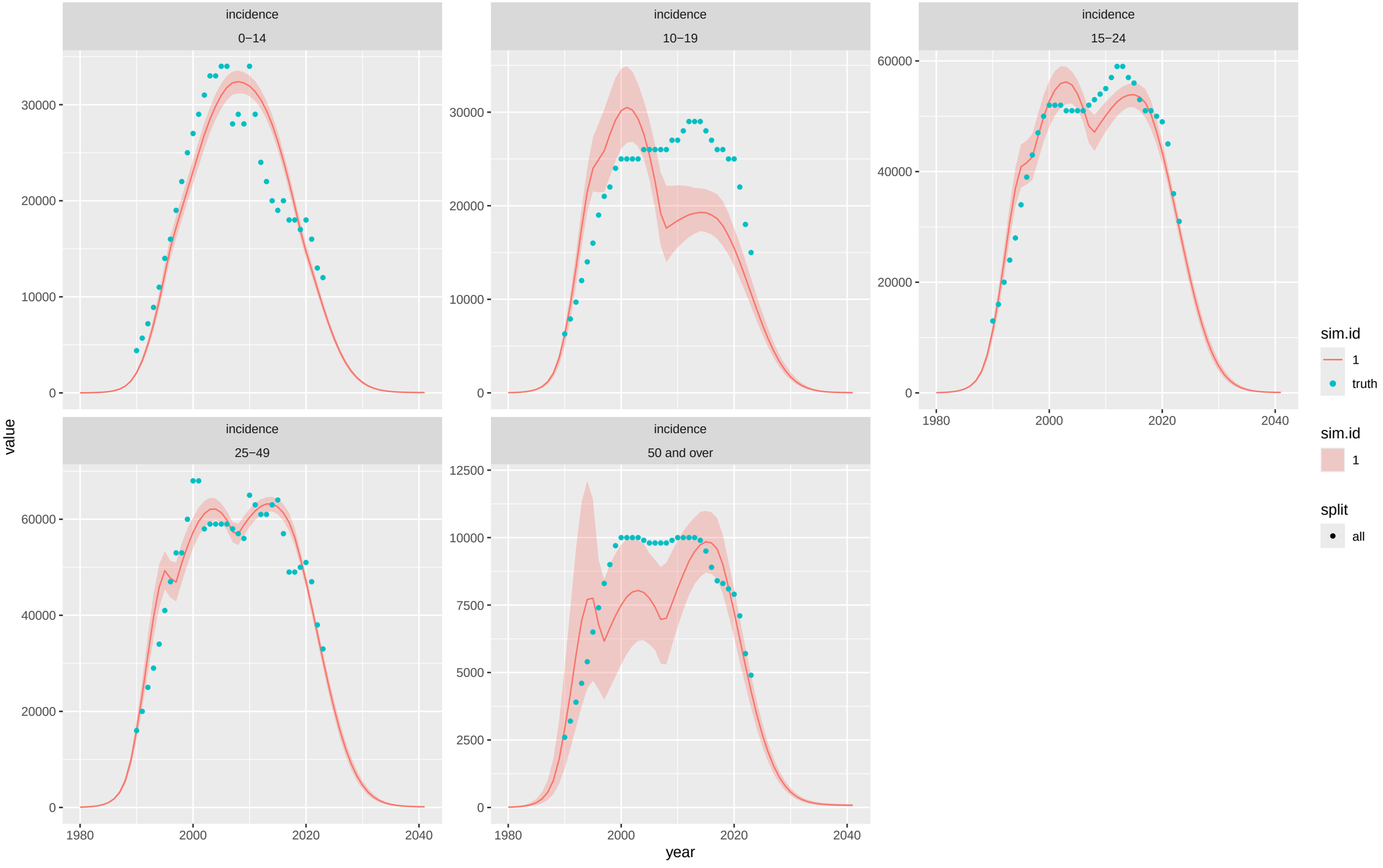
Mozambique step 2: cascade.by.age.sex (one color per chain)



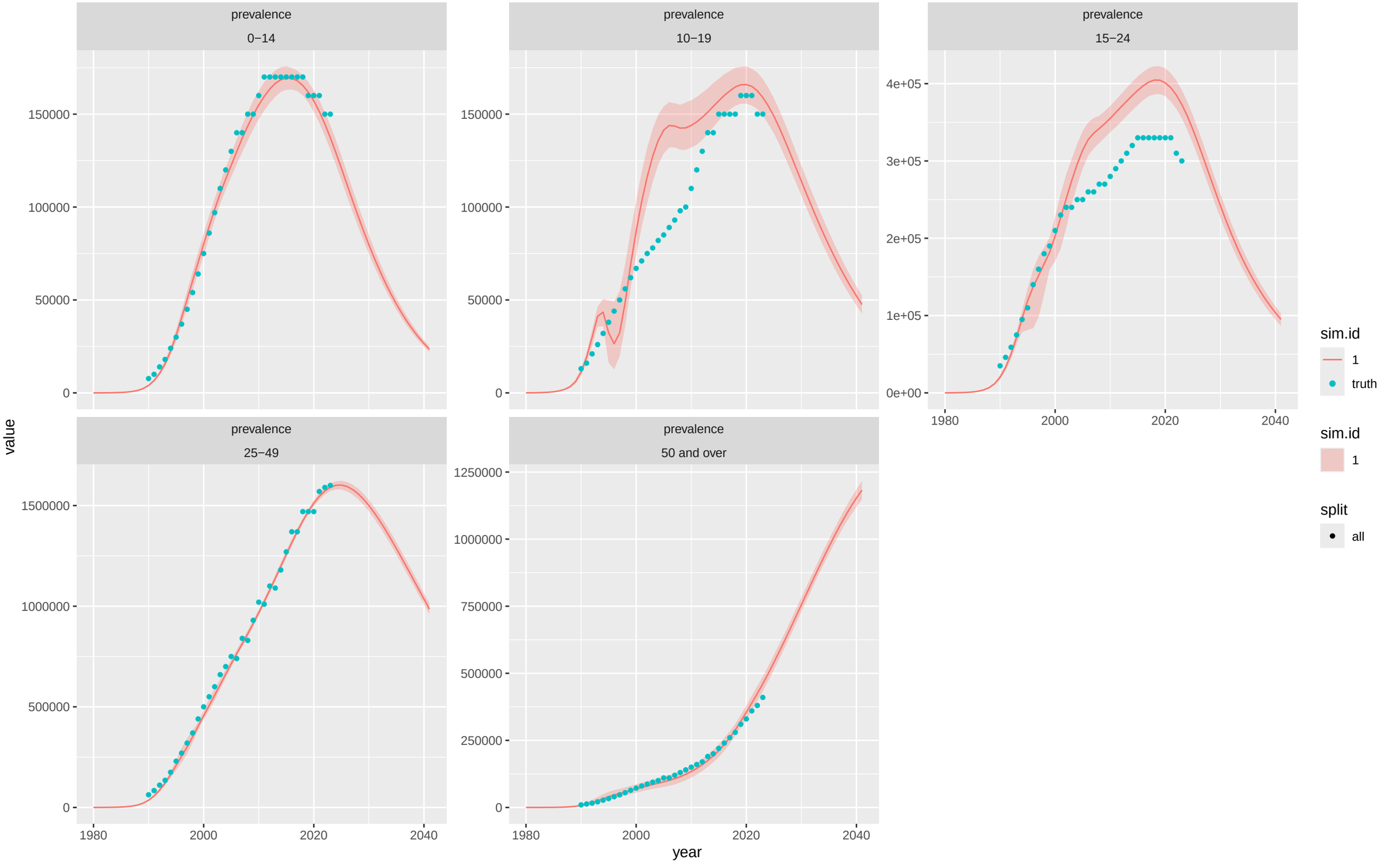
Mozambique: fit (merged Step 2 simset)



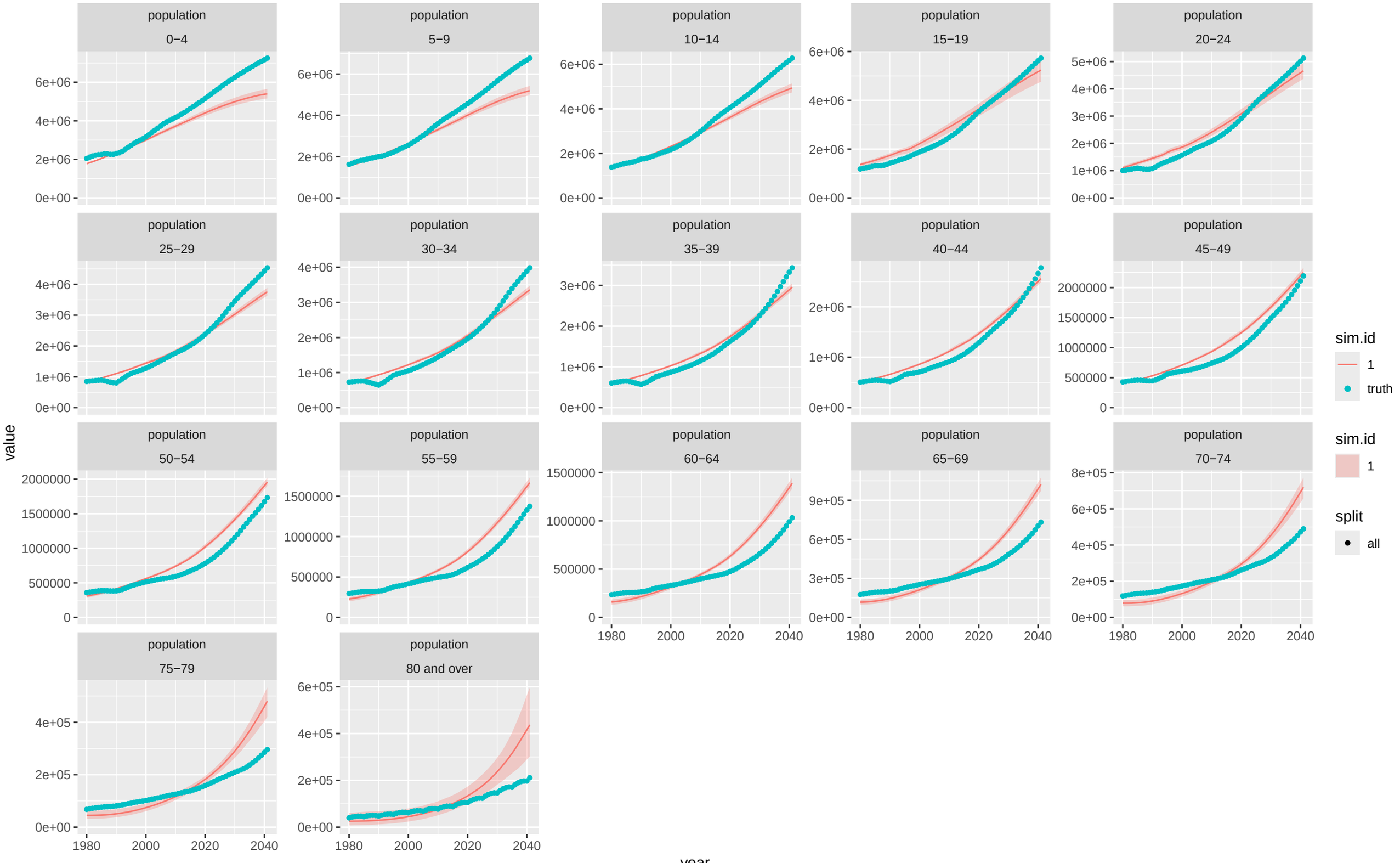
Mozambique: incidence.by.age (merged Step 2 simset)



Mozambique: prevalence.by.age (merged Step 2 simset)



Mozambique: population.by.age (merged Step 2 simset)



Mozambique: cascade (merged Step 2 simset)

awareness

engagement

suppression

value

1.00

0.75

0.50

0.25

0.00

1980

2000

2020

2040

year

sim.id

1

truth

sim.id

1

split

all

Mozambique: cascade.by.age.sex (merged Step 2 simset)

